ITMO

Literature Review.

Summarizing. Paraphrasing. Synthesis. Referring to sources.

3rd semester

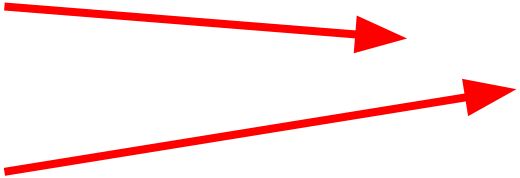
Literature Review Types

- Stand alone literature review as a single work
- Review paper
- Literature review as a part of a thesis / dissertation/ final year project
- Literature review as part of a research paper

Literature Review

Literature review

can be part of
Introduction
or a separate section of
a research paper



Title
Abstract
keywords
Introduction
Methods
Results
Discussion
Conclusion
Acknowledgement
Reference list

Literature Review Aims



Steps for Literature Analysis

- Define your research question
- Select the databases
- Specify the scope of your search (e.g. publication dates)
- Survey literature sources
- Analyse the information
- Synthesize the information
- Present the literature in an organized way

Literature Review Matrix

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Literature review helps to arrange the information from different sources in a systematic way, compare and contrast various aspects important for your research, find differences and similarities, pluses and minuses of research methods.

The information may be arranged in the matrix form.

Literature review matrix template

Literature Review Matrix

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Customize the Literature review matrix to **your** research needs.

Highlight the most important, relevant points for your research. Add, delete, rename columns to meet your research needs.

Literature review matrix template

Literature Review Matrix

The first part of the Literature review matrix usually contains the publication data of a source/paper and keywords.

	A	B	C	D	E	F	G	H	I	J
1	**Author(s) last name, first initial**	**Publication date**	**Title of article or chapter**	**Book title**	**Journal title**	**Database**	**Library**	**URL**	**DOI**	**Keywords**

Literature Review Matrix

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Literature review matrix should be tailored to your research needs and feature the most relevant information for your research, for example compare and contrast various methods and/or specify their pluses and minuses.

K	L	M	N	O
Method	**Design**	**Sample**	**Problem**	**Purpose**

Example

K	L	M
Method	Purpose of the method	Analysis (strengths and weaknesses compared to other studies)
Deterministic/ Partial derivatives	Calculating the exact slopes provided by the partial derivative of the model response y=f(x) with respect to the i-th model parameters xi	Pros: efficient for simple cost functions; Cons: explore only a small fraction of the design space, especially wher many parameters;
Indirect (or bruteforce) method	To find the divisors of a natural number n would enumerate all integers from 1 to n, and check whether each of them divides n without remainder.	Pros: simple to use and requires no additional model development; Cons: expensive for models based on nonlinear partial equations with many involves a trial-anderror process when selecting the parameter perturb
Direct	The sensitivity coefficients are calculated from an auxiliary set of equations derived from the model equations	Pros: for other applications, the auxiliary equations have been decoupled fro Cons: unstable or has failed completely for several stiff problems;

Literature Review Matrix

The matrix should feature the research question of an article, analyse strengths and weaknesses of the research under study, summarize the key points. Citation/reference list column provides the in-text and/or reference list citation in the style required for the publication of your research in the journal. You may want to use different citation columns for different citation styles (APA, MLA, IEEE, etc.).

P	Q	R	S
Research question	**Summary (what is the article about?)**	**Analysis (strengths and weaknesses compared to other studies)**	**Citation/Reference list entry**

Practice

Create the literature review matrix based on at least 2 sources.

Skills for Literature Analysis

Paraphrasing and synthesis are the skills that will help you to advance your Literature Review writing.

Paraphrasing

In Scientific writing **paraphrasing** refers to a process, when other person's idea or a piece of text is rewritten/said in your own words without changing the original meaning but with substantial structure changes and with proper reference/in-text citation.

Paraphrasing

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A slightly altered text that keeps similar syntax is **patchwriting**. Such texts, even with proper references, are considered **plagiarism**.

Original:

Similarly, LCs are known for their fascinating ability of controlling polarization and intensity of light, with plethora of technological applications, such as the flat panel displays, but LC-enabled motors or machines capable of converting external sources of energy into mechanical work are rarely demonstrated. (Yuan et al., 2018)

Paraphrased:

Likewise, LC's are known for their fascinating ability to control polarization and light brightness, with a plethora of technological applications, such as flat panel displays. LC motors or machines that can convert external energy sources into mechanical works are rarely identified. (Yuan et al., 2018)

Steps for Paraphrasing

- analyze the source text
- keep the terms and specific words
- express the information differently
    - change the word class
    - use synonyms
    - change voice and/or sentence structure
- rewrite the text
- reread the text
    - check that the meaning is the same
    - revise the text for coherence and cohesion
- use proper reference

Checklist for Paraphrasing

- ❑ meaning of the source text and the paraphrase is the same
- ❑ wording is different
- ❑ sentence structure differs significantly
- ❑ references are given

Summarizing

A **summary** is a short, clear description that gives the main facts or ideas about something.

dictionary.cambridge.org

Usually a summary is much shorter than the source text, as it presents the outline of ideas given in the original text.

Steps for Summarizing

- read the text
- identify the main idea of the text
- find keywords and terms
- formulate key ideas of the text in the note form
- write a short version (summary) of the text using your notes and keywords/terms
- reread the text for coherence and cohesion
- compare the relevance of ideas in the text written with the source text
- use proper reference

Checklist for Summary

- ❑ meaning of the source text and the summary is the same.
- ❑ wording is different, apart from keywords and terms.
- ❑ sentence structure differs significantly.
- ❑ summary is much shorter than the source text.
- ❑ references are given

SYNTHESIS

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- combines elements of several sources to help you make a point
- describes how sources converse each other
- helps to organize similar ideas together so readers can understand how they overlap
- helps readers see where you add your own new ideas to the existing knowledge

Steps for Synthesizing

- read and analyze the sources
- formulate key ideas for each source (2-3 ideas)
- critically analyze the texts asking: How do the ideas relate to the ideas in other texts on the same topic? Are they in agreement? Do the ideas differ in their perspective/scope/methods/etc. ?
- find the ideas that repeat in two or more sources and group them together to create arguments, and subarguments for your research question
- support your arguments by including specific subarguments that prove your point
- include the examples and details as supporting evidence for these arguments
- draw your conclusions on the state of the literature on the topic
- write the text using your notes
- reread the text for coherence and cohesion
- compare the relevance of ideas in the text written with the source texts
- use proper reference

SYNTHESIS MATRIX

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Organize the ideas using the matrix

Arranging your sources by theme or variable helps to identify how the sources relate to each other, and how to weave them together to create a narrative.

Research Question:

SYNTHESIS MATRIX	Source 1 Author:	Source 2 Author:	Source 3 Author:	Source 4 Author:
Argument 1:	Supporting evidence	Supporting evidence	Supporting evidence	Supporting evidence
Subargument:				
Subargument:				
Argument 2:	Supporting evidence	Supporting evidence	Supporting evidence	Supporting evidence
Subargument:				
Subargument:				

What is your conclusion? (The answer to your research question.)

LITERATURE REVIEW

Literature review is

- descriptive
- interpretative
- responds with critique
- demonstrates the author's voice

LANGUAGE TIPS

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Use **signposting words** to highlight your comparison of different sources

However, … Nevertheless, … In addition,… Similarly, …. Likewise,… In particular, …, while …, whereas…, in the same way …

Three key studies found that weight loss interventions that lasted one month in length were effective (Main, 2017; Major; Raj, 2018). **However,** studies that lasted six months showed that while participants lost weight initially, their weight rebounded by six months (Lee, 2008; Talmun, 2011; Harrington, 2013). These studies show that **while** short-term weight loss is possible, long term weight maintenance is challenging.

LANGUAGE TIPS

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Use **report verbs**

Steinke et al. [11] **proposed** a feature localization system for autonomous vehicles that leverages LiDAR and geometric fingerprinting to associate features.

Brown et al (2020) **ignore** the role of visual representation of online resources in students' online engagement.

Tyagi and Aswathy **explored** the open challenges in the smart autonomous vehicle. They **did not focus** on a specific stage or specific category of algorithms. However, they **described** the tasks and challenges in the ADS [20].

LANGUAGE TIPS

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Find critique and author's voice

These differences are of concern in the retention and success of PhD students from clinical backgrounds, as many of these attributes have been identified as risk factors for attrition. For example, the likelihood of successful completion has been shown to be higher for those students who begin their PhDs earlier, such as in their twenties, than those who enter at later ages. Moreover, completion rates decline as the age at inception increases (Martin, Maclachlan & Karmel, 2001; Wamala, Oonyu & Ocaya, 2011). This may be due to increased financial and personal responsibilities as students get older, making it more difficult to establish an appropriate work-life balance (Martin et al., 2001). Similarly, mature aged students have been shown to be less likely to participate in postgraduate student societies and other peer support activities, which may reduce their sense of belonging and increase their likelihood of attrition (Gardner, 2009; Govendi Sinns, Symons & Tammen, 2009; Humphrey & Simpson, 2012; Kehrhahn, Sheckley & Travers, 2000). These studies have provided insights into the experience of PhD students of various disciplinary backgrounds, but little has been reported regarding the PhD experience of students from clinical backgrounds. Therefore, it is difficult to hypothesise whether these factors are also significant for clinical PhD students, as they are for the more widely studied "mainstream" cohort.

LANGUAGE TIPS

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Find critique and author's voice

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Different Reference Style

Nowadays, rotation modulation technology is the primary method to realize the selfcompensation of errors in the SINS, which can suppress the bias drift and other slowly changing errors of inertial sensors by periodic rotation [13,14]. The inertial sensors are rotated around one axis or multiple axes [15–17]. The uniaxial rotation modulated inertial navigation system (RMINS) has been widely studied and applied [18].

In order to better understand the shape-recognition capabilities of dolphin biosonar with an aim to replicate it in a biomimetic system, we conduct EV-MTS experiments in a pool[8,9,10]

According to French et al. (2010), 38% of the fresh weight of processed onion is unfit for human consumption and is thrown away as waste. 10% of the moist weight of the onion is made up of dry outer skins and stems. It is estimated that about 10% of harvested onions are worthless marketed, meaning that they do not meet the marketing requirements for quality in any way. In the onion industry, there are a number of biological and environmental issues associated with the waste (onion skins), which amount to more than 550.000 tons per year (Chadorshabi et al., 2022).

Citation style

Citation style depends on the journal.

Check journal requirements for publication or guide for authors before submitting the manuscript.

Citation style manuals can also be found here

Find the journal for publication. Go to the journal website. Study the referencing requirements.

Practice

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Write the literature review part of the research paper.

References

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THANK YOU FOR YOUR TIME!

ITMO more than a UNIVERSITY